



DECODED: KEY AVIATION TECHNOLOGIES

Despite all the risks and hazards, here's how technology powers aviation to be among the safest ways of commercial travel across the world.

Souvik Das | souvik@digit.in

The field of aviation has always played guest to the latest advanced technologies in both commercial and combat guise - radars, augmented reality displays, you name it. But, there's a lot at stake here too. Take the Airbus A320. On a standard flying route between Mumbai to Delhi, it typically flies with over 150 people onboard, covering an aerial distance of 1,150 km in roughly 105 minutes cruising at a peak of around 35,000 ft, and assuming all conditions are ideal (no turbulent winds, optimal cruising speed, standard overall weight, etc.), will fly at around 850 kmph (ground speed), and consume around ₹2,15,000 worth of Jet A1 fuel, barring all other ancillary expenses.

Also, air travel, for obvious reasons, is riskier than road travel. Any instance of breakdown can be fatal. Yet, year-by-year, with significantly rising commuters, it produces the least number of deaths and injuries in comparison to road, or even train travel. The answer to this lies in the core technologies behind aviation. Neelu Khatri, President of Aerospace at Honeywell India, provides us the key details.



The Airbus A320neo - a new breed of ultra high-tech, safe aircrafts

TACKLING FLIGHT EFFICIENCY

Honeywell Aerospace has been building technologies for the aviation sector for nearly a century now. At Honeywell India, the key focus is on getting the Indian aerospace players onboard to use technologies that it develops for global service providers. "If we look at technology that is helping the India market specifically, it's worth thinking about the impact of rising oil prices, which is putting the Indian low-fare airline market under pressure to offset fuel costs by raising passenger airfares," Khatri says, "That's where Honeywell comes in. Our GoDirect Flight Efficiency software provides data analysis, reporting and monitoring tools to conserve fuel, while reducing costs and environmental impact."

The GoDirect flight efficiency software is a suite of algorithms onboard aircrafts and air terminals that help an airline become more efficient. These algorithms are run through simulators to train them on real-life situations, and are steadily improved using